

Multiple Choice (10 marks)

1. What is the value of the expression $2,816 \times 7$?
 - (a) 19,812
 - (b) 20,112
 - (c) 19,612
 - (d) 14,572
 - (e) 18,712
2. Suppose $x - 3$ and $y + 3$ are multiples of 7. What is the smallest positive integer, n , for which $x^2 + xy + y^2 + n$ is a multiple of 7?
 - (a) 9
 - (b) 8
 - (c) 5
 - (d) 6
 - (e) 3
3. If four points are picked independently at random inside the triangle ABC, what is the probability that no one of them lies inside the triangle formed by the other three?
 - (a) 0.1250
 - (b) 0.6667
 - (c) 0.4444
 - (d) 0.8000
 - (e) 0.5000
4. What is the greatest common factor of 36 and 90?
 - (a) 30
 - (b) 12
 - (c) 45
 - (d) 15
 - (e) 6
5. Find the last 3 digits of $2003^{(2002^{2001})}$.
 - (a) 806
 - (b) 001

- (c) 615
- (d) 404
- (e) 923

True / False (10 marks)

Answer True or False

6. True or False: The solution to the equation $v - 26 = 68$ is $v = 84$.
7. True or False: The Binormal vector for the given vector function $\mathbf{r}(t)$ is $[0.7, 0.0, 0.7]$.
8. True or False: If $g(x) = f(-x)$ and $g(3) = 2$, then $f(-2)$ must equal 3.
9. True or False: If a sum is invested at a continuous annual return rate of 7%, it will double in approximately 9.90 years.
10. True or False: Values such as 0.0042 and 0.005 are also acceptable choices for x within the specified range.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Find the curve defined by the equation

$$\theta = \frac{\pi}{3}.$$

(A) Line (B) Circle (C) Parabola (D) Ellipse (E) Hyperbola Enter the letter of the correct option.

12. Convert 813_9 to base 3.

End of Paper